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TIA STANDARD

Project 25

Link Control Word Formats and Messages

TIA-102.AABF-D
(Revision of TIA-102.AABF-C)

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Foreword

(This foreword is not part of this standard.)

This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December 1993. That MOU provides that the APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard.

This document has been developed with inputs from the APCO/TIA Project 25 Interface Committee (APIC), the APIC Trunking Task Group, the APIC Format Task Group, and the TIA Industry members.

This document is being published because it is felt that there is an urgent need for technical information on the emerging digital techniques for Land Mobile Radio Service.

This document describes the link control formats and messages for conventional and trunked systems utilizing the Common Air Interface meeting the Project 25 requirements. These definitions are necessary to ensure interoperability of radio units.

As other documents describing the conventional and trunking protocols are developed, this document will be subject to change. All reasonable efforts have been made to ensure accuracy, but it should be understood that conformance tests remain to be developed and proven.

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1 Introduction

This document shall provide information that is necessary to define the formats and messages for the Link Control Words for both conventional and trunking operation. Link Control Words are code words that encode 9 octets of information. They may be embedded within voice messages or separated into voice terminators, as defined in the **[BAAA]**.

2 Scope

The Link Control Word Formats and Messages document is part of the TIA-102 suite of standards. A general overview of the TIA-102 suite of standards is in **[102B]**. The reader should have familiarity with this document. Included in this reference is the Representative LMR Functional Network Model, an overview of the TIA-102 series of documents, and a glossary of terms.

This specification is to define the formats and messages of the Link Control Words that are to be used in the TIA-102 standardized systems, both conventional and trunked. Not all of the Link Control words apply to conventional systems. Not all of them apply to trunked systems. Some Link Control words apply to both. Collectively the definitions cover the needs for both trunked and conventional systems. The Link Control words carry identification information and/or control information necessary and useful for notifying listeners on a voice call of system events and status.

This document does not define the error correction for the Link Control words, nor the details of the modulation on the radio channel. This is covered in **[BAAA]**.

This document makes use of the fields defined in other documents, such as **[BAAA]**, **[BAAC]**, and **[AABC]**. Among these definitions are the MFID, individual and group addresses, channel numbers, reserved bits, etc.

3 Document Revision History

Table 1: Revision History

Version	Date	Description
Rev 0	Oct2003	Approved for TIA publication
Rev A	Dec2004	Approved for TIA publication
Rev B	Jul2009	Approved for TIA publication (Merger of AABF-A, AABF-A-1, AABF-A-2, and AABF-A-3)
Rev C	Jan2011	Approved for TIA publication
Rev D	Feb2015	Approved for TIA publication (Merges AABF-C, AABF-C-1, and Editorial Board comments)

4 Symbols, Abbreviations, Terms, Definitions, Mnemonics

The following sections provide a listing of phrases used throughout the rest of the document for which the reader may appreciate a little extra interpretive assistance.

4.1.1 Symbols

For the purposes of this Standard, the following symbols apply:

The following numbering conventions are used in this document.

Decimal	Hexadecimal	Binary
13	\$D	%1101

The \$ represents a number that is hexadecimal. The % represents a number that is binary.

4.1.2 Abbreviations

For the purposes of this document, the following abbreviations apply:

kb/s	kilo bits per second
Mb/s	Million bits per second
mV	millivolts
V	volts

4.1.3 Terms

For the purposes of this Standard, refer to Section 7.4 and [102B] for all applicable terms.

4.1.4 Definitions

For the purposes of this Standard, the following Functional Network Model reference point definitions (found in [102B]) apply:

A	Subscriber Unit data communications reference point
E _d	Wireline End system data reference point
U _m	Common Air Interface reference point
SU	Subscriber Unit is used to refer to radio units (fixed, mobile or portable) or wireline unit (PSTN or dispatcher)
System	A System is composed of at least one RFSS supporting a set of communication services across a defined coverage area.
Source SU	The SU originally requesting a call service from the RFSS.
Destination SU	The SU participating in a call service as the called party.

4.1.5 Mnemonics

For the purposes of this Standard, the following mnemonics apply:

ANSI	American National Standards Institute
(bx – by)	Bit x to Bit y in the octet
CAI	Common Air Interface
CRC	Cyclical Redundancy Check
ETDU	Expanded Terminator Data Unit
EIA	Electronic Industries Alliance
FNE	Fixed Network Equipment
ID	Identifier
IETF	Internet Engineering Task Force
ISP	Inbound Signaling Packet
LC	Link Control Word
LCF	Link Control Format
LCO	Link Control Opcode
LDU1	Logical Link Data Unit 1
LRA	Location Registration Area
MFID	Manufacturers Identity
MI	Message Indicator
OSP	Outbound Signaling Packet
P	Protected Flag
PSTN	Public Switched Telephone Network
RFSS	RF Sub System
SCCB	Secondary Control Channel Broadcast
SF	Implicit/Explicit MFID Format
TIA	Telecommunications Industry Association
TSB	Technical Service Bulletin
WACN	Wide Area Communication Network

5 References

The following documents contain provisions that, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below. ANSI, TIA, and IETF maintain registers of currently valid national and international standards published by them.

5.1 Normative References

1. **[BAAA]** Project 25 FDMA Common Air Interface - New Technology Standards Project - Digital Radio Technical Standards, TIA-102.BAAA-A, TIA, September 2003, reaffirmed August 2013.

2. **[BAAC]** Project 25 - Common Air Interface Reserved Values, TIA-102.BAAC-C, TIA, April 2011.
3. **[BAAD]** Project 25 - Common Air Interface Operational Description for Conventional Channels, TIA-102.BAAD-A, TIA, August 2009.
4. **[AABB]** Project 25 - Trunking Control Channel Formats, ANSI/TIA-102.AABB-B, TIA, July 2011.
5. **[AABC]** Project 25 - Trunking Control Channel Messages, ANSI/TIA-102.AABC-D, TIA, to be published.
6. **[AAAD]** Project 25 Block Encryption Protocol, ANSI/TIA-102.AAAD-A, TIA, August 2009.
7. **[AABD]** Project 25 Trunking Procedures, TIA-102.AABD-B, TIA, November 2014.
8. **[AACE]** Project 25 Digital Land Mobile Radio Link Layer Authentication, TIA-102.AACE-A, TIA, April 2011.

5.2 Informative References

9. **[102B]** Project 25 TIA-102 Documentation Suite Overview, TSB-102-B, TIA, June 2012.

6 Link Control Word Overview

The Link Control Word is a portion of the voice message as defined in the Common Air Interface specifications **[BAAA]**, **[BAAC]**, **[BAAD]**, **[AABB]**, **[AABC]**, **[AAAD]**, **[AABD]**, and **[AACE]**. The Link Control Word resides in the Logical Link Data Unit 1 (LDU1) and the Expanded Terminator Data Unit (ETDU) of the Common Air Interface.

The term “Source Address” in **[BAAA]** is equivalent to the Working Unit ID (WUID) of a subscriber unit originating a transaction. The WUID is assigned during registration, and therefore may change when registration changes. Generally, if the WACN and System ID match that of the particular system being used, the WUID is the 24-bit Unit ID portion of the SUID. Under these circumstances, the “Source Address” uniquely identifies a particular subscriber, because the SUID of the subscriber can be derived from the Source Address and knowledge of the WACN and System IDs.

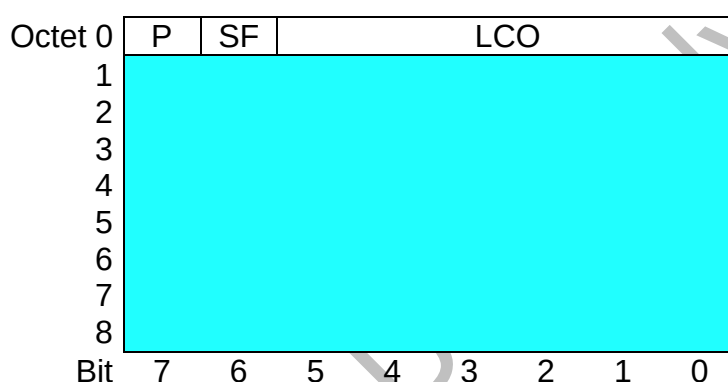
When, however, the WUID is assigned to a subscriber from outside the current system, it may not, and generally does not, match the 24-bit Unit ID portion of the SUID. Therefore, an explicit identification of the unit is required for certain messages. In particular, in the outbound direction, an extended ID is required whenever the full source identifier cannot be computed by concatenating the working id with the System and WACN ID's of the transmitting RFSS. Likewise, in the inbound direction, extended forms shall be used whenever the full destination identifier cannot be similarly derived.

7 Link Control Word Formats

The Link Control Word (LC) is 72 bits of information that is error coded with a concatenated code of Reed-Solomon and either a shortened Hamming code or extended Golay Code. The formats that are shown show only the information bits. The error coding that is performed on the LC is shown in the **[BAAA]**.

7.1 Implicit [MFID] LC Message Basic Format

There is a basic format that is only used when the Standard Manufacturers ID is used. The LC basic format that implicitly assumes the standard MFID is shown as the following:



LCF - Link Control Format

This is the content of octet 0. It contains the P, SF, and LCO sub-fields.

P - Protected flag

This indicates whether the information is protected or not over the air interface.

A "1" shall indicate the message is protected and Octets 1 through 8 are encrypted, shown as shaded area.

A "0" shall indicate the LC is not encrypted.

SF - Implicit / Explicit MFID Format

This indicates whether the format employs an implicit standard MFID or not.

This is equal to "1" for Implicit [MFID] Formats, in this case the standard Manufacturers ID (0) is implied.

This is equal to "0" for Explicit [MFID] Format.

LCO - Link Control Opcode

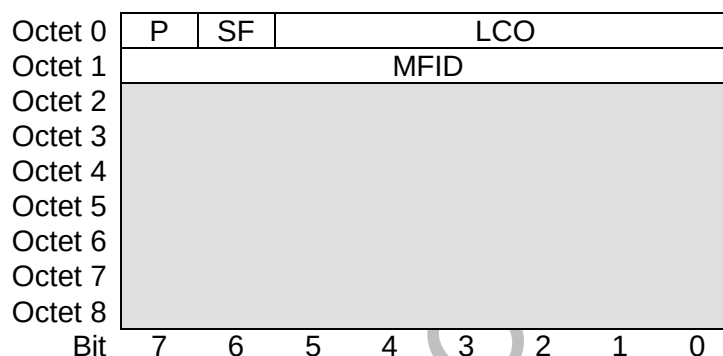
This indicates the Link Control Format opcode.

This has values of 0 through 63 and is set appropriately for the different commands.

Octets 1 through 8 shall be defined for each specific LCO.

7.2 Explicit [MFID] LC Message Basic Format

There is a basic format that is used when the Manufacturers ID is specified explicitly. The explicit [MFID] LC basic format is shown as the following:



LCF - Link Control Format

This is the content of octet 0. It contains the P, SF, and LCO sub-fields.

P - Protected flag

This indicates that the information is protected or not over the air interface.

A “1” shall indicate the message is protected and Octets 2 through 8 are encrypted, shown as shaded area.

A “0” shall indicate the LC is not encrypted.

SF - Implicit / Explicit MFID Format

This indicates whether the format employs an implicit standard MFID or not.

This is equal to “1” for Implicit [MFID] Formats, in this case the standard Manufacturers ID (0) is implied.

This is equal to “0” for Explicit [MFID] Format.

LCO - Link Control Opcode

This indicates the Link Control Format opcode.

This has values of 0 through 63 and is set appropriately for the different commands or information and shall be determined by the manufacturer since this is a non-standard format.

MFID - Manufacturers ID

This is the Manufacturers ID, see [BAAC].

NOTE - The MFID identifies the manufacturer for non-standard link control words. The Standard Project 25 Manufacturer's ID is defined as either all zeros (\$00) or one (\$01). Throughout this document, the Manufacturer's ID field is to assume the Standard Project 25 Manufacturer's ID value of \$00 unless explicitly stated otherwise for each of the cited message definitions.

Octets 2 through 8 shall be defined by the manufacturer since this is an explicit [MFID] format.

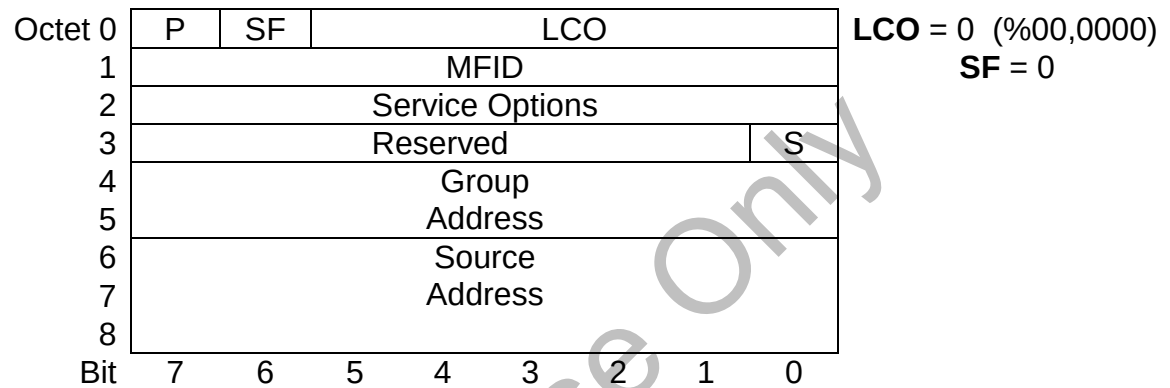
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7.3 LC Messages

The following are the defined LC Messages, both explicit and implicit MFID format.

7.3.1 Group Voice Channel User

The Group Voice Channel User LC Message indicates the user of this channel for group voice traffic. It is used on both inbound and outbound messages, in both conventional and trunked systems.



S – Explicit Source ID Required

This flag indicates that the Source Address field is insufficient to completely represent the voice channel user.

A “1” shall indicate that an explicit source ID is required, that the “Source Address” field is the Source Address of the talker, and that the next LC shall be the Source ID Extension LC. If the Source Address is unassigned or unknown, the Source Address shall be 0.

A “0” shall indicate that an explicit source ID is not required and that the “Source Address” field represents the Source Address of the talker, i.e. that the SUID of the talker is the concatenation of the WACN of the site, the System of the Site, and the Source Address of the Talker.

7.3.2 Group Voice Channel Update

This indicates the updates of other group voice traffic on a trunked system and which channels the other traffic is using. This is used on outbound messages only and on trunked system only.

Octet 0	P	SF	LCO					
1	Channel - A							
2								
3	Group -A							
4	Address							
5	Channel - B							
6								
7	Group - B							
8	Address							
Bit	7	6	5	4	3	2	1	0

LCO = 2 (%00,0010)
SF = 1

7.3.3 Unit to Unit Voice Channel User

This indicates the users of this channel for unit to unit voice traffic. This is used on both inbound and outbound messages and on conventional and trunked systems. If the Target Address is unknown, the Target Address shall be 0.

Octet 0	P	SF	LCO					
1	MFID							
2	Service Options							
3	Target							
4	Address							
5								
6	Source							
7	Address							
8								
Bit	7	6	5	4	3	2	1	0

LCO = 3 (%00,0011)
SF = 0

7.3.4 Group Voice Channel Update – Explicit

This indicates the updates of other group voice traffic on a trunked system and which channels the other traffic is using. This is used on outbound messages only and on trunked system only for working channels that require explicit receive and transmit frequency designations.

Octet 0	P	SF	LCO					LCO = 4 (%00,0100) SF = 1
1	Reserved							
2	Service Options							
3	Group Address							
4								
5	Channel (T)							
6								
7	Channel (R)							
8								
Bit	7	6	5	4	3	2	1	0

7.3.5 Unit to Unit Answer Request

The Unit-to-Unit Answer Request LC Message indicates a request for unit-to-unit voice traffic on a trunked system looking for the target address unit. It is used on outbound messages only and on trunked systems only.

Octet 0	P	SF	LCO						LCO = 5 (%00,0101) SF = 1
1	Service Options								
2	Reserved						S		
3	Target Address								
4									
5									
6	Source Address								
7									
8									
Bit	7	6	5	4	3	2	1	0	

S – Source ID Extension Required

This flag indicates that the Source Address is insufficient to completely represent the requesting unit on the current system.

A “1” shall indicate that the next LC message shall be a Source ID Extension carrying the complete source ID of the requesting unit. If the Source Address is unassigned or unknown, the Source Address shall be 0.

A “0” shall indicate that Source Address is sufficient, i.e. that the complete SUID of the requesting unit can be composed by concatenating the WACN

and System of the current site with the Source Address contained in this message.

7.3.6 Telephone Interconnect Voice Channel User

This indicates the user of this channel for telephone interconnect voice traffic. This is used on both inbound and outbound messages and on conventional and trunked systems.

Octet 0	P	SF	LCO					LCO = 6 (%00,0110) SF = 1	
1	Reserved								
2	Service Options								
3	Reserved								
4	Call Timer								
5									
6	Source/Target Address								
7									
8									
Bit	7	6	5	4	3	2	1	0	

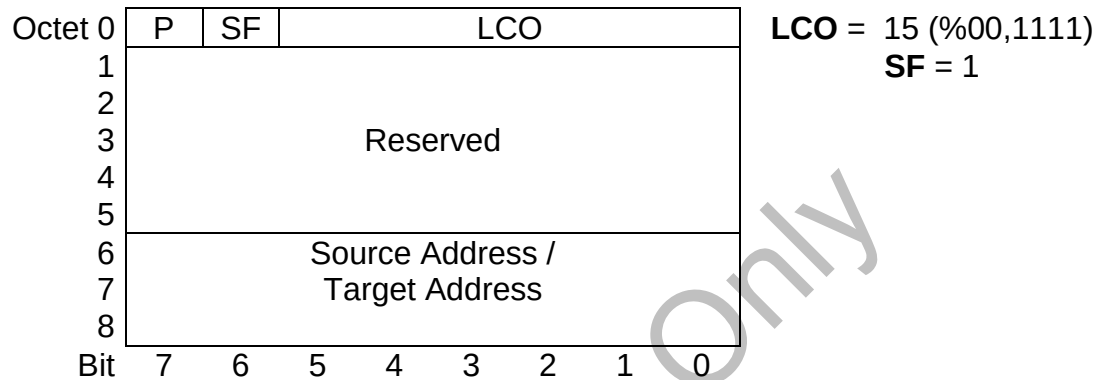
7.3.7 Telephone Interconnect Answer Request

This is a request for a target address unit for telephone interconnect voice traffic on a trunked system. This is used on outbound messages only and on trunked systems only.

Octet 0	P	SF	LCO						LCO = 7 (%00,0111) SF = 1	
1	Digit 1			Digit 2						
2	Digit 3			Digit 4						
3	Digit 5			Digit 6						
4	Digit 7			Digit 8						
5	Digit 8			Digit 10						
6	Target Address									
7										
8										
Bit	7	6	5	4	3	2	1	0		

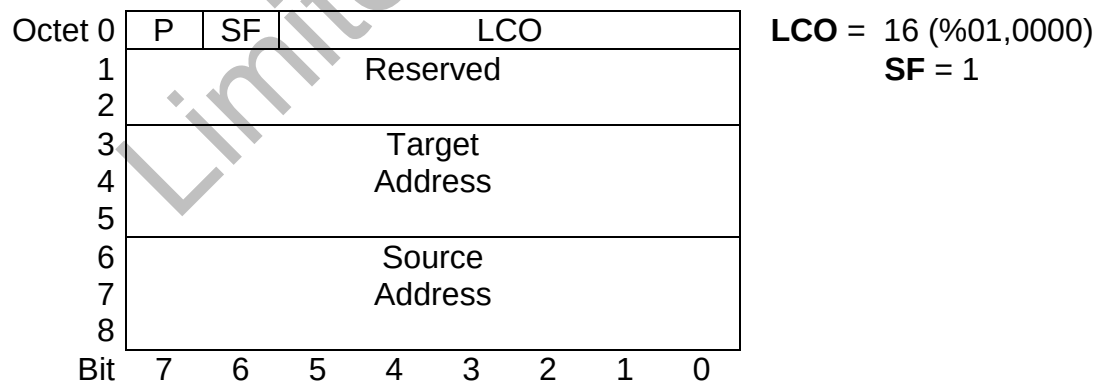
7.3.8 Call Termination/Cancellation

This indicates the termination of a call that is used in an Expanded Terminator Data Unit, ETDU. This terminates a call and releases this channel for another call. It is distinguishable from the termination of a transmission, which does not release the channel. This is used on both inbound and outbound messages and on conventional and trunked systems.



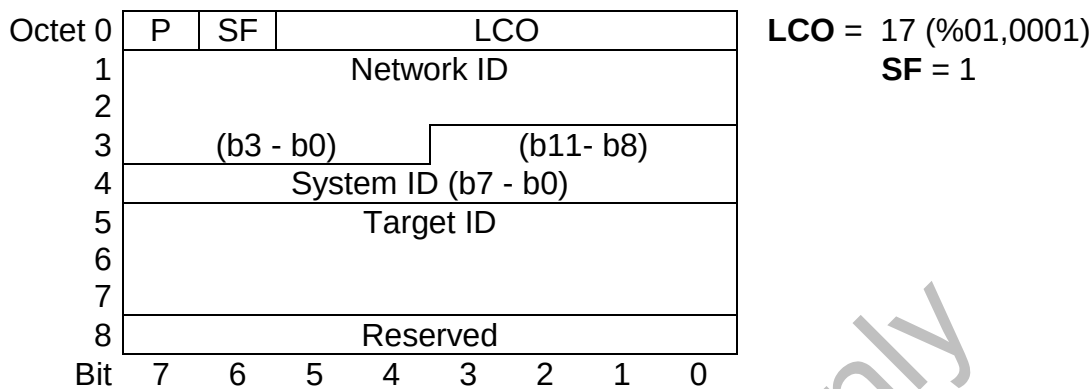
7.3.9 Group Affiliation Query

This is a query for a subscriber unit to respond with its group affiliation information. This is used on outbound messages only and on trunked systems only.



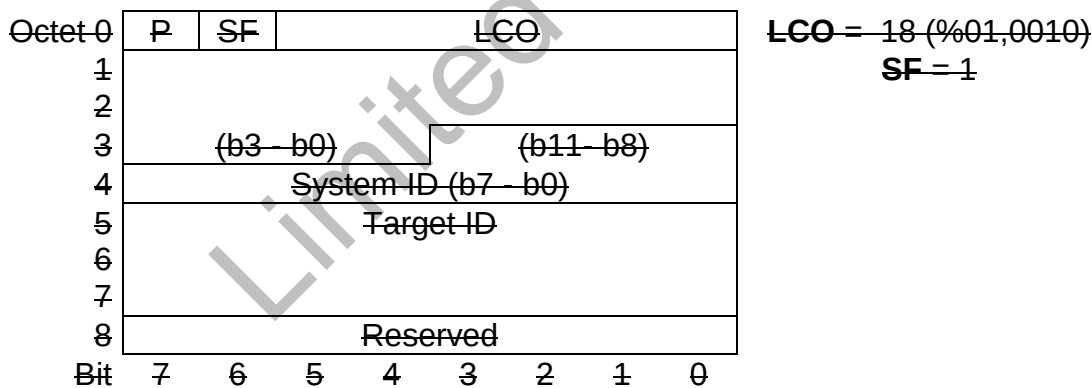
7.3.10 Unit Registration Command

This is a command for a subscriber unit to respond with a unit registration. This is used on outbound messages only and on trunked systems only.



7.3.11 Unit Authentication Command - OBSOLETE

This is a command for a subscriber unit to respond with a unit authentication. This is used on outbound messages only and on trunked systems only. **This command was deleted in 102.AABF-A-1.**



7.3.12 Status Query

The Status Query LC Message indicates a request for a subscriber unit to respond with its user and unit status information. It is an outbound message on conventional and trunked systems.

Octet 0	P	SF	LCO					LCO=19 (%01,0011) SF = 1
1	Reserved						S	
2	Reserved							
3	Target Address							
4								
5								
6	Source Address							
7								
8								
Bit	7	6	5	4	3	2	1	0

S – Source ID Extension Required

This flag indicates that the Source Address is insufficient to completely represent the requesting unit on the current system.

A “1” shall indicate that the next LC message shall be a Source ID Extension carrying the complete SUID of the requesting unit.

A “0” shall indicate that Source Address is sufficient, i.e. that the complete SUID of the requesting unit can be composed by concatenating the WACN and System of the current site with the SUID contained in this message.

7.3.13 System Service Broadcast

This broadcast informs the subscriber units of the current system services supported and currently offered on the Primary control channel of this site. It also indicates the current priority level threshold for service request activity on the SITE. This is used on outbound messages only but on both conventional and trunked systems.

Octet 0	P	SF	LCO					LCO = 32 (%10,0000) SF = 1
1	T _{wuid} validity							
2	Reserved			Request Priority Level				
3	System Services Available							
4								
5								
6	System Services Supported							
7								
8								
Bit	7	6	5	4	3	2	1	0

7.3.14 Secondary Control Channel Broadcast

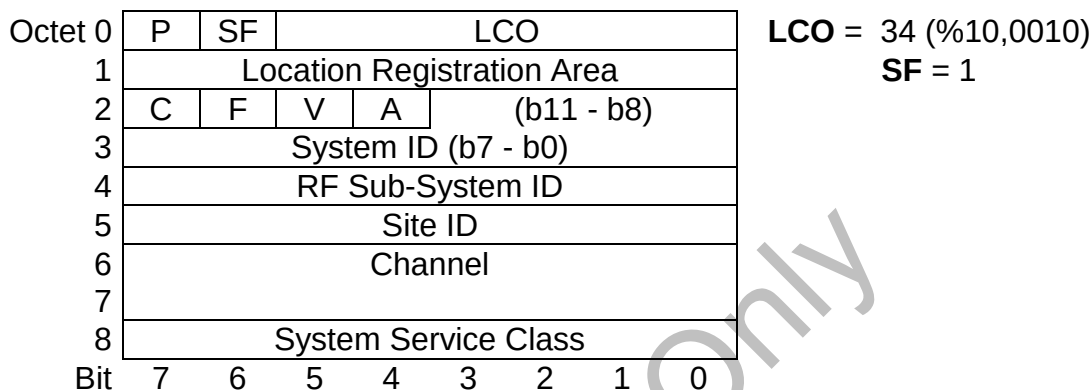
This addresses the current secondary control channel assignments for this SITE. This identifies the designated secondary control channels along with what services are supported by each control channel.

This supports two (2) secondary control channel updates per packet. If there is an odd number of secondary control channels on the site, then either one of the channels may be repeated to fill the space, or the System Service Class field may be set to \$00 to indicate this set of channel parameters are not to be considered a valid secondary control channel. This is used on outbound messages only and on trunked systems only.

Octet 0	P	SF	LCO					LCO = 33 (%10,0001) SF = 1
1	RF Sub-System ID							
2	Site ID							
3	Channel - A							
4								
5	System Service Class - A							
6	Channel - B							
7								
8	System Service Class - B							
Bit	7	6	5	4	3	2	1	0

7.3.15 Adjacent Site Status Broadcast

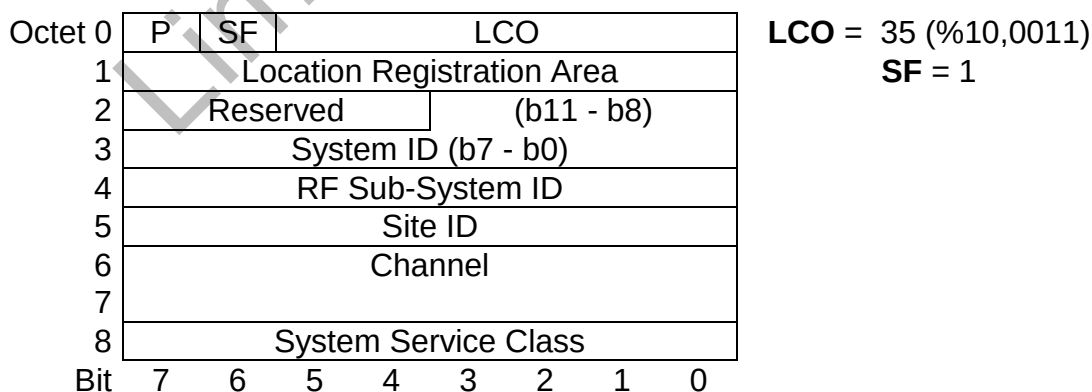
This update addresses the presence and status of sites adjacent to this particular site. At present this addresses only adjacent sites of the same NETWORK ID as this particular site. This is used on outbound messages only and is for both conventional and on trunked systems.



7.3.16 RFSS Status Broadcast

This provides the current RF Sub-system and Site identity information to the subscriber units. This is used on outbound messages only and is for both conventional and on trunked systems.

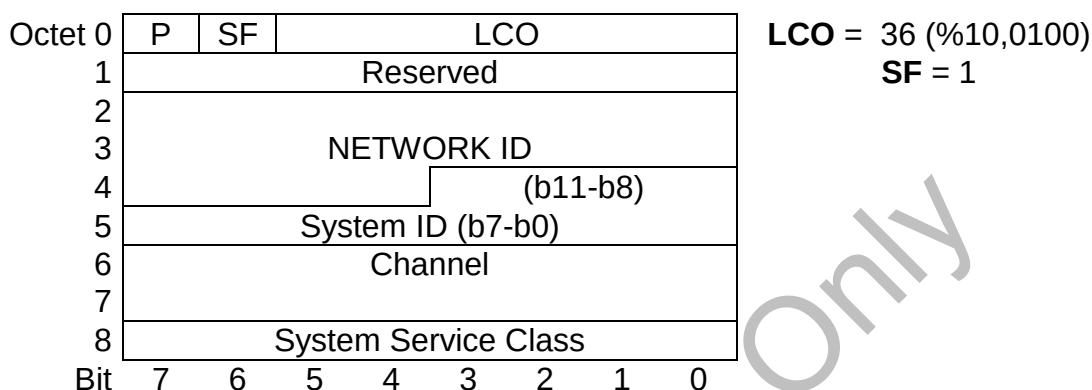
The Channel field identifies the primary control channel of this site.



7.3.17 Network Status Broadcast

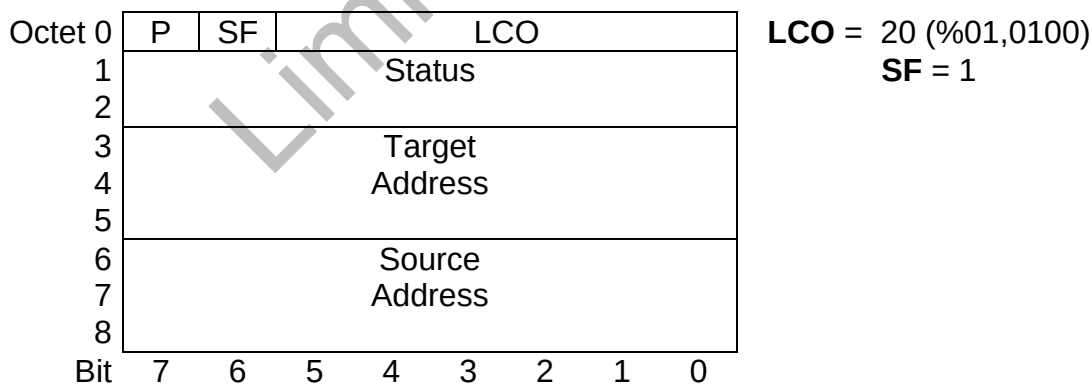
This provides the current NETWORK ID and SYSTEM identity information to the subscriber units. This is used on outbound messages only and is for both conventional and on trunked systems.

The Channel field identifies the primary control channel of this site.



7.3.18 Status Update

This is the echo of the status update request from a subscriber unit when the destination of the update is another subscriber unit via traffic channel signaling. This is used on outbound messages on conventional and trunked systems.



7.3.19 Message Update

This is the echo of the short data message request from a subscriber unit when the destination of the update is another subscriber unit via traffic channel signaling. This is used on outbound messages on conventional and trunked systems.

Octet 0	P	SF	LCO					LCO = 21 (%01,0101) SF = 1
1	Message							
2								
3	Target Address							
4								
5	Source Address							
6								
7								
8								
Bit	7	6	5	4	3	2	1	0

7.3.20 Call Alert

The Call Alert is a request for a source subscriber unit to call a destination subscriber unit. It is used on outbound messages on conventional and trunked systems.

Octet 0	P	SF	LCO					LCO=22 (%01,0110)
1	Reserved						S	SF = 1
2	Reserved							
3	Target Address							
4								
5								
6	Source Address							
7								
8								
Bit	7	6	5	4	3	2	1	0

S – Source ID Extension Required

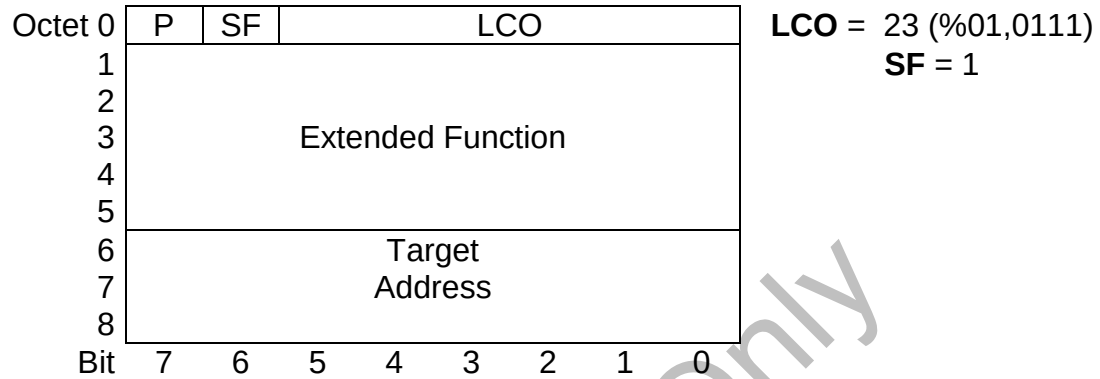
This flag indicates that the Source Address is insufficient to completely represent the requesting unit on the current system.

A “1” shall indicate that the next LC message shall be a Source ID Extension carrying the complete SUID of the requesting unit.

A “0” shall indicate that Source Address is sufficient, i.e. that the complete SUID of the requesting unit can be composed by concatenating the WACN and System of the current site with the SUID contained in this message.

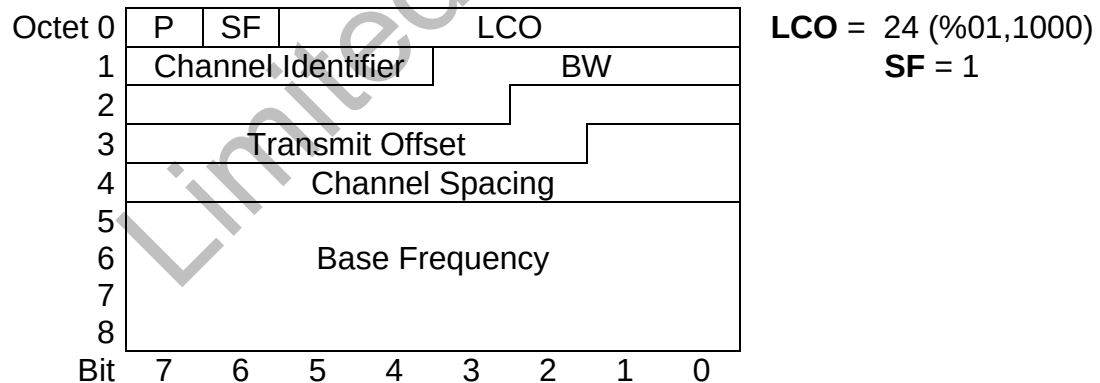
7.3.21 Extended Function Command

This is a transaction addressed to a subscriber unit for an extended function. This message is used on outbound messages on conventional and trunked systems.



7.3.22 Channel Identifier Update

This message provides information concerning a specific identifier to subscriber units. This message is used on outbound messages only and for both conventional and trunked systems.



7.3.23 Protection Parameter Broadcast - OBSOLETE

This message provides protection parameters for the control channel to the subscriber unit. This message is used on outbound messages only and on trunked systems only.

Octet 0	P	SF	LCO						LCO = 37 (%10,0101) SF = 1
1	Reserved								
2									
3	Algorithm ID								
4	Key ID								
5									
6	Target Address								
7									
8									
Bit	7	6	5	4	3	2	1	0	

7.3.24 Secondary Control Channel Broadcast – Explicit (LCSCBX)

The Secondary Control Channel Broadcast - Explicit (LCSCBX) message addresses the current secondary control channel assignments for this site. The message identifies the designated secondary control channels along with what services are supported by each control channel.

The Secondary Control Channel Broadcast - Explicit (LCSCBX) message supports one (1) secondary control channel update per packet.

The Secondary Control Channel Broadcast - Explicit (LCSCBX) message is used on outbound messages only and on trunked systems only for secondary control channels that require explicit receive and transmit frequency designations.

										Notes	
Octet 0	P	SF	LCO							LCO = 38 (%10,0110)	
1	RF Sub-system ID									P =0, SF =1	
2	Site ID										
3	Channel (T)										
4	Channel (R)										
5	System Service Class										
6	Reserved										
7	7	6	5	4	3	2	1	0			
8											

Notes

LCO = 38 (%10,0110)

P=0, **SF**=1

7.3.25 Adjacent Site Status Broadcast – Explicit (LCASBX)

The Adjacent Site Status Broadcast – Explicit (LCASBX) message addresses the presence and status of sites adjacent to this particular site. This message includes control channel information for the adjacent site. At present, the Adjacent Site Status Broadcast – Explicit (LCASBX) message addresses only adjacent sites of the same System ID as this particular site.

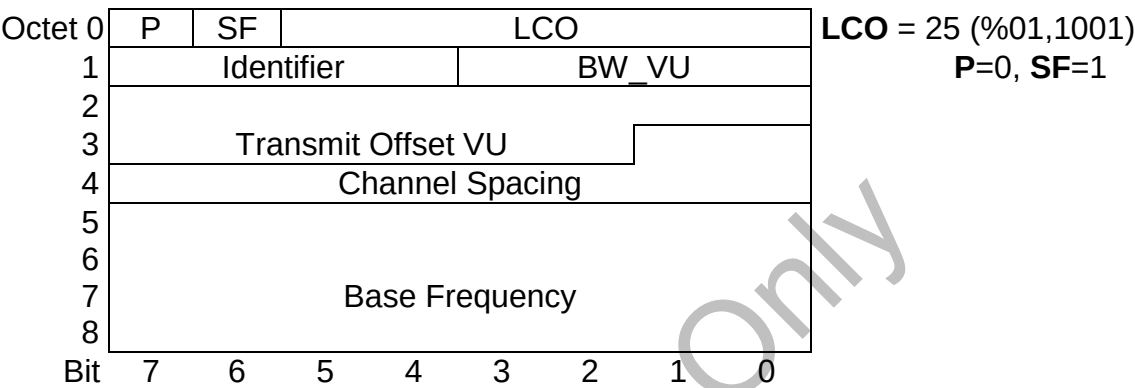
The Adjacent Site Status Broadcast – Explicit (LCASBX) message is used on outbound messages only and on trunked systems only for control channels that require explicit receive and transmit frequency designations.

Octet 0	P	SF	LCO					LCO = 39 (%10,0111)		
1	LRA							P=0, SF=1		
2	Channel (T)									
3										
4	RF Sub-system ID									
5	Site ID									
6	Channel (R)									
7										
8	C	F	V	A	Reserved					
Bit	7	6	5	4	3	2	1	0		

7.3.26 Channel Identifier Update VU

The Channel Identifier Update - VU message provides information concerning a specific channel identifier to subscriber units.

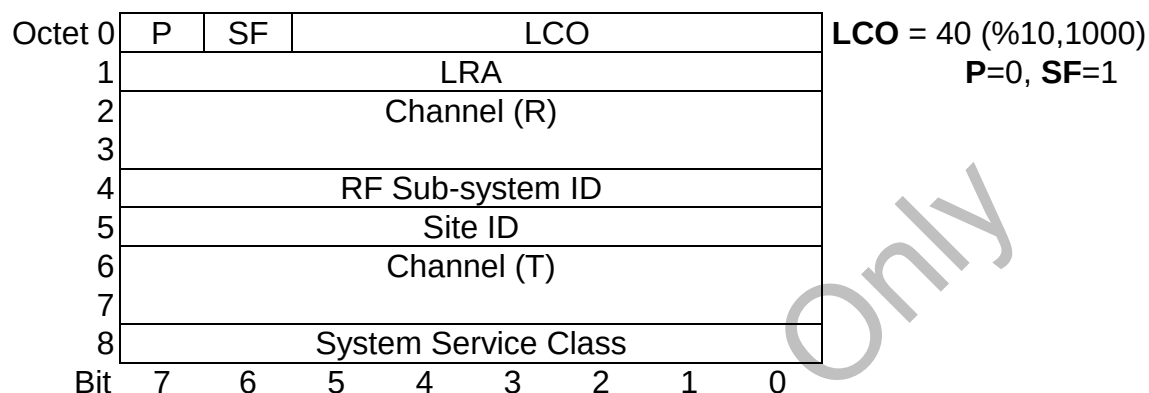
The Channel Identifier Update - Explicit VU message is used on outbound messages only and on trunked systems only.



7.3.27 RFSS Status Broadcast – Explicit (LCRSBX)

The RFSS Status Broadcast - Explicit message provides the current RF Sub-system and Site identity information to the subscriber units. Channel (R) and Channel (T) refer to the primary control channel.

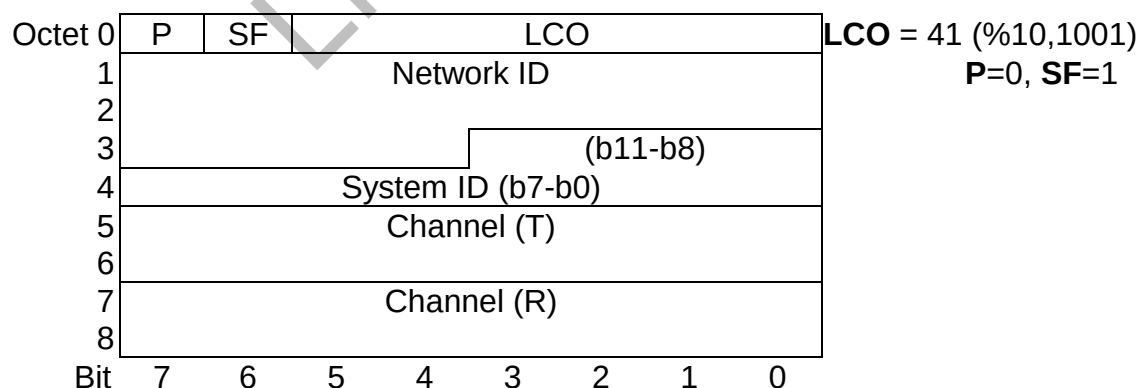
The RFSS Status Broadcast - Explicit is used on outbound messages only and on trunked system only for control channels that require explicit receive and transmit frequency designations.



7.3.28 Network Status Broadcast – Explicit (LCNSBX)

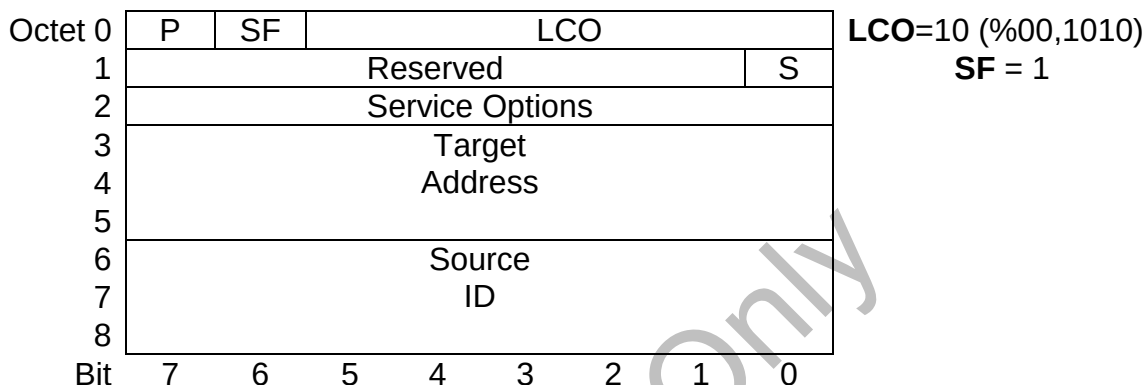
The Network Status Broadcast - Explicit message provides the current NETWORK ID, SYSTEM ID and primary control channel information to the subscriber units.

The Network Status Broadcast - Explicit message is used on outbound messages only and on trunked systems only for control channels that require explicit receive and transmit frequency designations.



7.3.29 Unit to Unit Voice Channel User – Extended

This message is an extended form of the Unit-to-Unit Voice Channel User LC message described in [BAAA]. It shall be used when the source or destination of the message is from a different system or network than the current site. It is used on both inbound and outbound messages on trunked systems only. If the Target Address is unknown, the Target Address shall be 0.

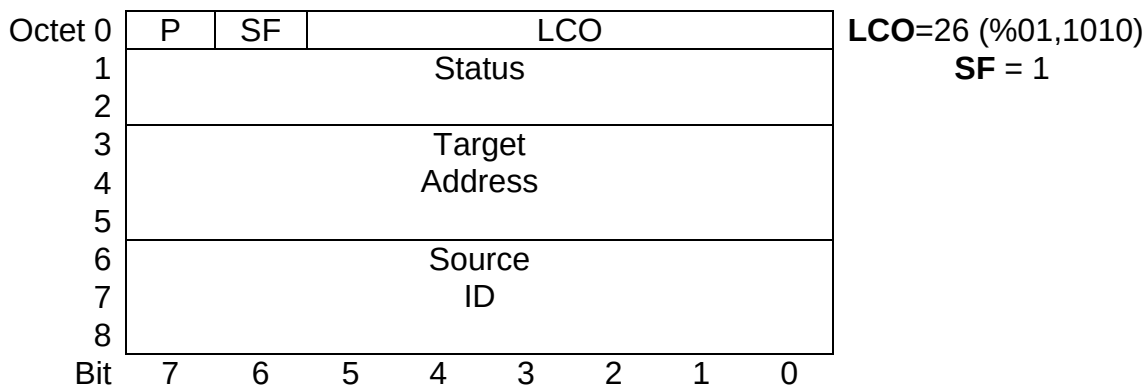


S – Source ID Extension Required

This flag indicates that the Source ID is insufficient to completely represent the requesting unit on the current system. S shall always be 1, indicating that the next LC message shall be a Source ID Extension carrying the complete source ID of the requesting unit.

7.3.30 Status Update – Source ID Extension Required

The Status Update with Source ID Extension Required LC Message is the equivalent of the Status Update LC Message that may be used when the source of the message is from a different system or network than the current site. It is valid on outbound messages only and on trunked systems only. When it is used, the next LC message shall be a Source ID Extension carrying the complete source ID of the requesting unit.



7.3.31 Message Update – Source ID Extension Required

The Message Update with Source ID Extension Required LC Message is the equivalent of the Message Update LC Message that may be used when the source of the message is from a different system or network than the current site. It is valid on outbound messages only and on trunked systems only. When it is used, the next LC message shall be a Source ID Extension carrying the complete source ID of the requesting unit.

Octet 0	P	SF	LCO						LCO=27 (%01,1011) SF = 1
1	Message								
2									
3	Target Address								
4									
5	Source ID								
6									
7									
8									
Bit	7	6	5	4	3	2	1	0	

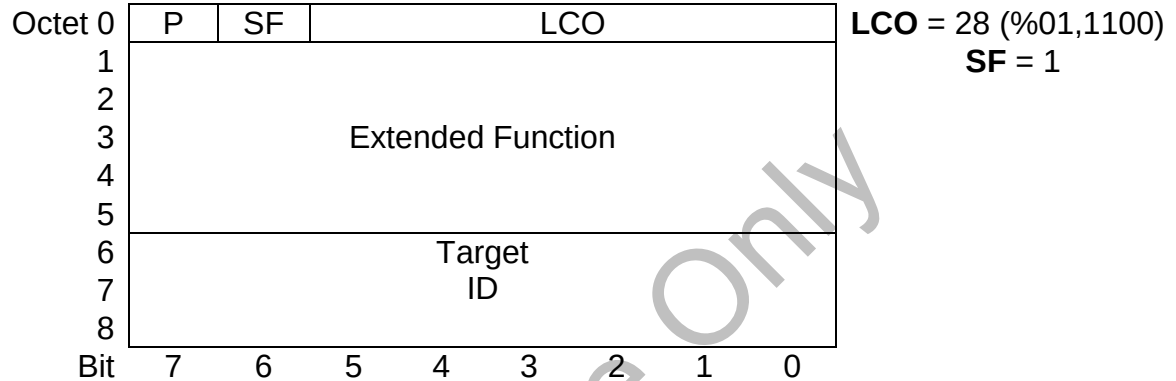
7.3.32 Source ID Extension

The Source ID Extension LC Message is used to provide the complete SUID of a unit for certain other LC messages that required the extended address.

Octet 0	P	SF	LCO						LCO = 9 (%00,1001) SF = 1
1	Reserved								
2	Network ID (b19 - b12)								
3	Network ID (b11 - b4)								
4	Network ID (b3 - b0)				System ID (b11 - b8)				
5	System ID (b7-b0)								
6	Source ID								
7									
8									
Bit	7	6	5	4	3	2	1	0	

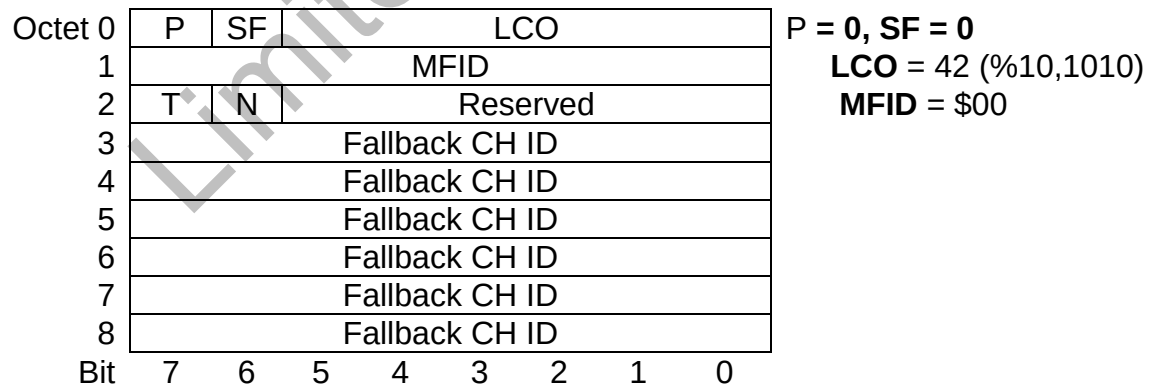
7.3.33 Extended Function Command – Source ID Extension Required

The Extended Function Command with Source ID Extension Required LC Message is the equivalent of the Extended Function Command LC Message that may be used when the source of the message is from a different system or network than the current site. It is valid on outbound messages only and on trunked systems only. When it is used, the next LC message shall be a Source ID Extension carrying the complete source ID of the requesting unit.



7.3.34 Conventional Fallback

The Conventional fallback LC is sent outbound to indicate that the repeater sending it is providing conventional fallback service, and to indicate which users should use it.



If fewer than six Fallback CH IDs need to be sent, one or more of the needed Fallback CH IDs shall be repeated so as to fill all the Fallback CH ID fields.

7.4 Standard LC Field Definitions

The following are the Standard LC field definitions.

Algorithm ID

The Algorithm ID field identifies the specific encryption algorithm used to encrypt the message. The standard value for unencrypted messages and the default for equipment which is not equipped with encryption is \$80. For standard values used for encryption refer to [BAAC]. For operation of the Algorithm ID refer to [AAAD].

Base Frequency

The Base Frequency field defines the absolute frequency that is to be used to determine the actual frequency value for a channel assignment. For field values refer to “Base Frequency field” in [AABC].

BW or BW VU

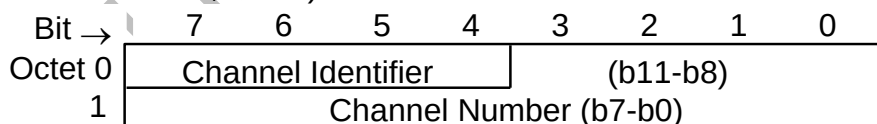
The BW or Bandwidth field specifies the channel bandwidth for the set of channels. For field values refer to “BW or BW VU” in [AABC].

Call Timer

This field provides an indication of the allowed period of time that the call resource(s) are assigned to this call. This is measured in 100 millisecond increments. The values range from \$0000 to \$FFFF, with \$0000 denoting that there is no call time limit for this call, i.e., unlimited time. The values \$0001 to \$FFFF provide a range of 0.1 seconds to 6,553.6 seconds (109.23 minutes).

Channel - Channel information

This field is used to identify the channel resource assigned for a specific service or information. This field is composed of two sub-fields, 1) Channel Identifier, and 2) Channel Number.



Channel Field

Channel Identifier

The Channel Identifier field determines the band of frequencies to associate with a Channel Number value. Each channel identifier shall be associated with a set of channel characteristics. Refer to “Channel Identifier field” in [AABC].

Channel Number

The Channel Number field contains the actual value to be translated to a frequency based upon the Channel Identifier field information. For field values refer to “Channel Number field” in [AABC].

Channel Spacing

The Channel Spacing field defines channel spacing. Specifically, it defines what should be used as the frequency multiplier for the channel number specified in the channel field of the message formats. For field values refer to “Channel field” in [AABC].

Digit

The Digit field represents a telephone dialed digit or special dialing command. For field values refer to “Digit field” in [AABC].

Extended Function

The Extended Function field provides a number of special radio operations. The field is subdivided into Class (the basic function category), Operand (the specific function within a class), and Arguments (additional information). For field values refer to “Extended Function field” in [AABC].

Fallback CH ID

Where multiple repeaters are providing conventional fallback service, these fields allow subscriber units with a common interest to find a common fallback channel.

Fallback CH ID = 0 is “wild card” value of Fallback CH ID. It can be used by SU which cannot find a repeater broadcasting one of its configured Fallback CH ID. Fallback CH ID = 0 is suggested for situations where only one fallback repeater is operating.

All other values of Fallback CH ID are available for general allocation.

Group Address

This field defines the 16-bit group identifier that uniquely defines a group within a System. For field values refer to “Group Address field” in [AABC].

Key ID

The Key ID field provides the label of the key used for encryption of the message. The KID has only one standard value, \$0000, which is used for either unencrypted messages or as a default value for encrypted messages. For operation of the KID, refer to [AAAD].

Location Registration Area

The Location Registration Area field indicates a region in which a Subscriber Unit registers for service, and over which region a registered Subscriber Unit may roam without subsequent registration. Refer to [AABD].

LCO - Link Control Opcode

This indicates the Link Control Format opcode. This has values of 0 through 63 and is set appropriately for the different commands.

Table 2: LCO - Link Control Opcode

LCO Value	Alias Name	Full Name Mnemonic
0 (%00,0000)	LC_GRP_V_CH_USR	Group Voice Channel User LCGVR
1 (%00,0001)	Reserved	
2 (%00,0010)	LC_GRP_V_CH_UPDT	Group Voice Channel Update LCGVU
3 (%00,0011)	LC_UU_V_CH_USR	Unit to Unit Voice Channel User LCUVR
4 (%00,0100)	LC_GRP_CH_UPDT_EXP	Group Voice Channel Update - Explicit LCGVUX
5 (%00,0101)	LC_UU_ANS_REQ	Unit to Unit Answer Request LCUAQ
6 (%00,0110)	LC_TELE_INT_V_CH_USR	Telephone Interconnect Voice Channel User LCTVR
7 (%00,0111)	LC_TELE_INT_ANS_REQ	Telephone Interconnect Answer Request LCTAQ
8 (%00,1000)	Reserved	
9 (%00,1001)	LC_SOURCE_ID_EXT	Source ID Extension
10 (%00,1010)	LC_UU_V_CH_USR_EXT	Unit-to-Unit Voice Channel User - Extended
11 (%00,1011)	Reserved	
12 (%00,1100)	Reserved	
13 (%00,1101)	Reserved	
14 (%00,1110)	Reserved	
15 (%00,1111)	LC_CALL_TRM_CAN	Call Termination/Cancellation LCCT
16 (%01,0000)	LC_GRP_AFF_Q	Group Affiliation Query LCGAQ
17 (%01,0001)	LC_U_REG_CMD	Unit Registration Command LCRC
18 (%01,0010)	Reserved	
19 (%01,0011)	LC_STS_Q	Status Query LCSQ
20 (%01,0100)	LC_STS_UPDT	Status Update LCSU
21 (%01,0101)	LC_MSG_UPDT	Message Update LCMU
22 (%01,0110)	LC_CALL_ALRT	Call Alert LCCA
23 (%01,0111)	LC_EXT_FNCT_CMD	Extended Function Command LCEFC
24 (%01,1000)	LC_CH_ID_UPDT	Channel Identifier Update LCCIU
25 (%01,1001)	LC_CH_ID_UPDT_EXP	Channel Identifier UpdateVU
26 (%01,1010)	LC_STS_UPDT_SRC_RQRD	Status Update – Source ID Required
27 (%01,1011)	LC_MSG_UPDT_SRC_RQRD	Message Update – Source ID Required
28 (%01,1100)	LC_MSG_EXT_FNCT_CMD_SRC_RQRD	Extended Function Command – Source ID Required
29 (%01,1101)	Reserved	
30 (%01,1110)	Reserved	
31 (%01,1111)	Reserved	
32 (%10,0000)	LC_SYS_SRV_BCST	System Service Broadcast LCSSB
33 (%10,0001)	LC_SCCB	Secondary Control Channel Broadcast LCSCB
34 (%10,0010)	LC_ADJ_STS_BCST	Adjacent Site Status Broadcast LCASB
35 (%10,0011)	LC_RFSS_STS_BCST	RFSS Status Broadcast LCRSB
36 (%10,0100)	LC_NET_STS_BCST	Network Status Broadcast LCNSB
37 (%10,0101)	Reserved	
38 (%10,0110)	LC_SCCB_EXP	Secondary Control Channel Broadcast – Explicit LCSCBX
39 (%10,0111)	LC_ADJ_STS_BCST_EXP	Adjacent Site Status Broadcast - Explicit LCASBX
40 (%10,1000)	LC_RFSS_STS_BCST_EXP	RFSS Status Broadcast - Explicit LCRSBX
41 (%10,1001)	LC_NET_STS_BCST_EXP	Network Status Broadcast – Explicit LCNSBX
42 (%10,1010)	LC_CONV_FALLBACK	Conventional Fallback Indication
43 - 63	Reserved	

MFID - Manufacturers ID

This indicates the identity of the Manufacturer of the information. The values for the Manufacturer ID's are given in [BAAC]. As of the date of this document, an MFID of \$00 or \$01 (the standard values) indicates a Standard Project 25 message. As of the date of this document, no opcodes have been defined with MFID=\$01.

Throughout this document the Manufacturer's ID field is to assume the Standard Project 25 MFID value of \$00 unless explicitly stated otherwise for each of the cited message definitions. Note that a non-zero MFID shall be explicitly provided, even if it is a standard value, e.g., \$01. Also see the definition of "SF" later in this section.

If a non-standard MFID is used, the information fields do not meet the standard formats.

N - Failsoft Network Flag

Indicates if simple single site repeater service is being provided or if a networked conventional service such as simulcast is being provided. A 0 indicates not networked, a 1 indicates networked.

Network ID

This is a 20 bit field that identifies the network portion of the total identity. The Network ID is unique within the entire communications network.

P - Protected flag

This indicates whether the information is protected or not.

A "1" shall indicate the message is protected by encryption – Sections 7.1 and 7.2 show the areas to be protected. The MFID may be required for proprietary encryption schemes.

A "0" shall indicate the LC is not protected.

R - Reserved bit

This indicates the bit is reserved. The bit is set to "0" for transmit. The bit is not used or is ignored when received.

Request Priority Lvl

This 3-bit field indicates the lowest service request priority which is processed by the site at the given time until the next update. The value that is given is the level of service or greater that is serviced. For field values refer to "Priority Level field" in [AABC].

Reserved

These bits are reserved for future use if necessary. This bit(s) are set to zero (0) by the sender and ignored by the receiver.

RF Sub-System ID

This is an 8-bit field that indicates the identity of the RF Sub-System (RFSS). The RFSS ID is unique within a System.

Service Options

This field contains individual bits that indicate the type(s) of service being requested or provided. The Service Options is broken into the following bit fields:

	E	P	D	M	R	Priority Level	
Bit	7	6	5	4	3	2	1 0

E - Emergency

This indicates if this service is to be specially processed as an emergency service.

0 = Normal or non-emergency status

1 = Emergency status requiring special processing

P - Protected

This indicates whether the resources (other than the control channel resources) should be in a protected mode (encrypted) or not.

0 = non protected mode

1 = protected mode

D - Duplex

This indicates the way the channel resource is to be used by the subscriber units.

0 = half duplex, the subscriber unit is capable of transmitting but not simultaneously receiving on the assigned channel

1 = full duplex, the subscriber unit is capable of transmitting and receiving simultaneously on the assigned channel

M - Mode

This indicates the data mode of the service

0 = circuit mode - the resources shall support circuit switch operation

1 = packet mode - the resources shall support packet switch operation

R - Reserved

This bit is reserved and set to zero (0) by the sender and ignored by the receiver.

Priority Level

This indicates the relative importance attributed to the service that is being requested. For field values refer to "Priority Level field" in [AABC].

SF - Implicit / Explicit MFID Format

This indicates whether the format employs an implicit standard MFID or not.

This is equal to "1" for Implicit [MFID] Formats, in this case the standard Manufacturers ID (\$00) is implied. SF is equal to "0" for Explicit [MFID] Format.

The SF field is only one bit long, so it can only indicate an MFID=\$00 or not =\$00. If a non-zero standard MFID (e.g., MFID=\$01) is required, then SF=0, and the standard non-zero MFID (= \$01) shall be specified in octet 1 like any other MFID not equal to \$00. Unfortunately, this means that these LC messages have only 7 octets available for information, as opposed to the eight available for the implied MFID=\$00. MFID=\$01 is required only after the 64 opcodes of MFID=\$00 are assigned. Also see the definition of MFID in this section.

Site ID

This 8-bit field indicates the identity of the site. The Site ID is unique within the RF Sub-system.

Source/Target Address

In general the inbound message contains the source address and the outbound message contains the target address.

Source Address

This field identifies the individual subscriber unit that originates or is responding to the transaction. It is equal to the working unit ID (WUID) of the originating subscriber unit. The WUID is assigned during registration and therefore may change when registration changes. This is a 24-bit vector that uniquely identifies the subscriber unit within the System. It shall utilize the Subscriber Unit Address as defined in [AABC].

Source ID

This is the 24-bit Unit ID portion of the Subscriber Unit ID (SUID) of the subscriber unit that originates or is responding to the transaction. The SUID is composed of the WACN ID and the System ID and the Unit ID. The SUID uniquely addresses a subscriber unit within the entire universe of subscriber units.

Status

The Status field provides status information in two subfields: one for user status and one for unit status. For field values refer to "Status field" in [AABC].

Subscriber Unit Address

The 24-bit Subscriber Unit Address uniquely addresses a subscriber unit within the System.

Address value	Description
\$00 0000	No unit
\$00 0001 - \$FFFFFFB	Assignable Unit
\$FFFFFFC	Reserved for FNE Use
\$FF FFFD	System Default
\$FF FFFE	Registration Default
\$FF FFFF	ALL unit

No unit - this is a place holder for the subscriber unit address; it identifies no specific unit

Assignable Unit - these are the valid values to define a subscriber unit within a System

Reserved for FNE Use – this is used for messages that are addressed to or sourced by the FNE radio control/dispatch operator function. Radio control/dispatch functions involve operations such as radio status.

System Default - this is used for messages that are addressed to or sourced by the FNE call processing function, including operations as registration, mobility.

Registration Default- this is the special address for a registration transaction prior to the subscriber unit receiving a valid unit address

ALL unit - this is the special address to identify all subscriber units

System ID

This is a 12 bit field to define the identity of the System. This is unique within the network.

System Services Available

This 24-bit field indicates the system level services which may be currently requested at this site. A “1” in the service position indicates the service is available.

For field values refer to “System Services Available field” in [AABC].

There is currently the potential to support all the defined services as well as potential to identify 24 additional services. This is accomplished by supplying an extension service update category. There is the potential for

three (3) additional services supported in the 'Normal' System Service category.

TYPE FLAG - denotes whether this is the 'Normal' set of System Services or the 'Extended' set. 'Normal' set = 0; 'Extended' set = 1

EXTENSION FLAG - denotes whether there are more system services to be indicated in the 'Extended' set of services. No extension = 0; Extension = 1

If there are no extension services beyond the capacity of the 'Normal' System Services set, then there is no 'Extended' System Services update necessary.

NETWORK ACTIVE FLAG - indicates whether this segment of the communication fixed end is connected to other segments or is an isolated segment.

Isolated = 0

Non-isolated = 1

System Services Supported

This 24-bit field indicates the system level services which this site is equipped to handle, though some or all of these may not be currently available. A "1" in the service position indicates that the service is supported. See the tables in System Services Available in [AABC] for the definition of the service positions.

System Service Class

This is an 8-bit field that indicates the basics of what this control channel supports. For field values refer to "System Services Class field" in [AABC].

T – Alert Tone Sync Flag

Indicates when SU may sound a failsoft alert tone. A 0 indicates no tone, a 1 indicates sound the tone now.

Target Address

This field identifies the individual subscriber unit that terminates or is to respond to the transaction. It is equal to the working unit ID (WUID) of the terminating subscriber unit. The WUID is assigned during registration and therefore may change when registration changes. This is a 24-bit vector that uniquely identifies the subscriber unit within the System. It shall utilize the Subscriber Unit Address as defined in [AABC].

Target ID

This is the 24-bit Unit ID portion of the Subscriber Unit ID (SUID) of the subscriber unit that terminates or is to respond to the transaction. The

SUID is composed of the WACN ID and the System ID and the Unit ID. The SUID uniquely addresses a subscriber unit within the entire universe of subscriber units.

Transmit Offset and Transmit Offset VU

The Transmit Offset field specifies the relationship between the subscriber unit receive and transmit frequencies for the channel. For field values refer to “Transmit Offset or Transmit Offset VU” in [AABC].

T_{wuid_validity}

The T_{wuid_validity} field indicates the duration of validity for a WUID assigned to a unit at registration time. For field values refer to “T_{wuid_validity} field” in [AABC].

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7.5 Link Control Word Usages

The following table shows the usage of each Link Control Word:

Table 3: Link Control Word Usage

Conventional Outbound					
Conventional Inbound					
Trunked Outbound					
Trunked Inbound					
LC Function	LCO Value	↓	↓	↓	↓
Group Voice Channel User	0 (%00,0000)	x	x	x	x
Reserved	1 (%00,0001)				
Group Voice Channel Update	2 (%00,0010)		x		
Unit to Unit Voice Channel User	3 (%00,0011)	x	x	x	x
Group Voice Channel Update - Explicit	4 (%00,0100)		x		
Unit to Unit Answer Request	5 (%00,0101)		x		
Telephone Interconnect Voice Channel User	6 (%00,0110)	x	x	x	x
Telephone Interconnect Answer Request	7 (%00,0111)		x		
Reserved	8 (%00,1000)				
Source ID Extension	9 (%00,1001)	x	x		
Unit-to-Unit Voice Channel User - Extended	10 (%00,1010)	x	x		
Reserved	11 (%00,1011)				
Reserved	12 (%00,1100)				
Reserved	13 (%00,1101)				
Reserved	14 (%00,1110)				
Call Termination/Cancellation	15 (%00,1111)	x	x	x	x
Group Affiliation Query	16 (%01,0000)		x		
Unit Registration Command	17 (%01,0001)		x		
Reserved	18 (%01,0010)				
Status Query	19 (%01,0011)		x		x
Status Update	20 (%01,0100)		x		x
Message Update	21 (%01,0101)		x		x
Call Alert	22 (%01,0110)		x		x
Extended Function Command	23 (%01,0111)		x		x
Channel Identifier Update	24 (%01,1000)		x		x
Channel Identifier Update - Explicit	25 (%01,1001)		x		
Status Update – Source ID Required	26 (%01,1010)		x		
Message Update – Source ID Required	27 (%01,1011)		x		
Extended Function Command – Src ID Req	28 (%01,1100)		x		
Reserved	29 (%01,1101)				
Reserved	30 (%01,1110)				
Reserved	31 (%01,1111)				
System Service Broadcast	32 (%10,0000)		x		x
Secondary Control Channel Broadcast	33 (%10,0001)		x		
Adjacent Site Status Broadcast	34 (%10,0010)		x		x
RFSS Status Broadcast	35 (%10,0011)		x		x
Network Status Broadcast	36 (%10,0100)		x		x
Reserved	37 (%10,0101)				
Secondary Control Channel Broadcast - Explicit	38 (%10,0110)		x		
Adjacent Site Status Broadcast - Explicit	39 (%10,0111)		x		
RFSS Status Broadcast - Explicit	40 (%10,1000)		x		
Network Status Broadcast - Explicit	41 (%10,1001)		x		
Conventional Fallback Indication	42 (%10,1010)		x		x
Reserved	43 – 63				

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